

# Synthetic-First Ordering Helps, but Not More Than Real Data:

## A Matched-Token Test of Two-Stage Synthetic Training in a Tiny Language Model

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Produced by an autonomous AI agent (Claude) on behalf of @pranavnandyalam. Not peer reviewed.

**Abstract**—A recent linear-regression analysis of SGD (Li and Zou, 2026) predicts that training on synthetic data first and real data second (S→R) avoids the error floor that mixing synthetic and real data incurs. We test the ordering prediction, and whether synthetic tokens are worth their training budget at all, in a 0.79M non-embedding-parameter GPT trained on TinyStories, with synthetic text sampled from a generator of the same architecture trained on a disjoint real split. With  $N$  real tokens and  $N$  synthetic tokens ( $N \in \{1M, 4M\}$ , 3 seeds, validation perplexity), S→R beats mixing at  $N=4M$  by 1.27 perplexity (95% hierarchical-bootstrap CI  $[-1.42, -1.11]$ , all seeds) and beats the reverse order by 2.99; at  $N=1M$  the ordering difference is within noise ( $-0.21$ , CI  $[-0.79, +0.22]$ ). The pre-registered criterion that the S→R advantage grows with  $N$  is met (contrast 1.06 perplexity, CI excluding zero), with only two values of  $N$ . However, against controls with matched total tokens, neither synthetic arm helps: at  $N=4M$ , training two epochs on the same  $N$  real tokens reaches  $15.18 \pm 0.14$  perplexity versus  $16.12 \pm 0.28$  for S→R and  $17.39 \pm 0.20$  for mixing; at  $N=1M$ , S→R, mixing and two-epoch real training are indistinguishable. The large gains of the synthetic arms over real-only training at  $N$  tokens are thus accounted for by the extra optimization steps. This is a small-scale negative result for the practical value of self-generated synthetic data, not a refutation of the theory, whose conditions our setup does not fully satisfy.

**Index Terms**—synthetic data, model collapse, curriculum ordering, small language models, data repetition, negative result

### I. INTRODUCTION

Language-model training increasingly mixes model-generated text into real corpora. A line of work on *model collapse* shows that training on generated data can degrade models [1], [2], [3], while accumulating real data alongside synthetic data can avoid collapse [4]. Li and Zou [5] analyze SGD in high-dimensional linear regression and conclude that mixing synthetic and real data leaves an error floor that does not vanish with more data, whereas “two-stage training avoids the floor by using synthetic data only in the first stage.” If this transferred to language models it would give a simple recipe: put synthetic data first.

Two questions matter for practice. First, does the order (synthetic first vs. mixed) change outcomes in an actual autoregressive transformer? Second, and more important for anyone deciding whether to generate synthetic data, does the

synthetic data beat simply spending the same training budget on the real data one already has? Comparisons that match only the number of real tokens conflate the value of synthetic data with the value of more optimization steps.

### Contributions.

- 1) A pre-registered, seed-paired test of synthetic-first (S→R) vs. mixed vs. reverse (R→S) ordering in a 0.79M-non-embedding-parameter GPT on TinyStories [6] at two real-token budgets, with a hierarchical (seeds, then documents) bootstrap. S→R beats mixing by 1.27 perplexity at  $N=4M$ ; at  $N=1M$  the difference is within noise.
- 2) Matched-total-token controls (fresh real tokens, and a second epoch over the same real tokens) showing that, in this setup, neither S→R nor mixing beats real-only training at the same number of steps; at  $N=4M$  two epochs of real data beat S→R by 0.94 perplexity.
- 3) A transparent account of what the experiment cannot show (two values of  $N$ , one synthetic fraction, a schedule that violates a theorem condition), together with code, raw per-run results and exact commands.

### II. RELATED WORK

**Theory of ordering.** Li and Zou [5] prove, for SGD in high-dimensional linear regression under distribution shift, that mixed training has an error floor and that two-stage training (synthetic, then real) avoids it, and give conditions under which synthetic pretraining beats real-only training. Their Section 5 reports a WikiText-103 language-model experiment with a pretrained teacher and states that it shows “the same qualitative behavior as predicted by our theory”; the details are in their Appendix H, which we could not retrieve (the arXiv HTML we fetched was truncated before it). Our work is therefore an independent test, not a replication of their LM setup. It differs in data (TinyStories), generator (a same-size model trained on a disjoint real split, not a stronger pretrained teacher), the inclusion of a reverse-order arm and a mixed-then-real-tail arm as recency controls, and, most importantly, matched-total-token real-only baselines. Marchi et al. [7] derive a minimum human-data fraction from a Fisher-Rao analysis; they do not study ordering.

**Model collapse.** Shumailov et al. [1] show that recursive training on generated data removes distribution tails; Dohmatob et al. [2] show that even a small synthetic fraction in a mixture can prevent gains from more data (mixed training only); Wang et al. [3] prove collapse for recursive training when real data is finite; Gerstgrasser et al. [4] contrast replacing vs. accumulating data across generations. We study a single generation and focus on ordering and budget accounting, not recursion.

**Synthetic data in LM pretraining.** Zhu et al. [8] report that training on self-generated data degrades LMs and propose token-level editing of human text to obtain semi-synthetic data. Kang et al. [9] study synthetic (rephrased) data in pretraining at scale; their abstract reports speedups from mixing about one third rephrased data with natural text and no advantage from rephrased data alone. Neither, to our reading of their abstracts (we have not read them in full), compares synthetic-first against mixed ordering at matched budgets in small models. Muennighoff et al. [10] find that, under a fixed compute budget, up to about four epochs of repeated data yield negligible loss changes compared with unique data; this motivates our two-epoch real-only control, which turns out to be the strongest baseline.

**Novelty statement.** A web search (2026-10-06/07; the Semantic Scholar API was rate-limited, so the check was not exhaustive) found no empirical ordering comparison of this kind on small LMs with matched-total-token controls. We claim no more than that.

### III. METHOD

#### A. Data splits and synthetic corpus

We split deduplicated TinyStories documents into four disjoint pools:  $R_{\text{gen}}$  (trains the generator),  $R_{\text{train}}$  (the real pool for all arms),  $R_{\text{dev}}$  (learning-rate selection only) and  $R_{\text{val}}$  (final evaluation only). A generator G0 with the same architecture is trained on  $R_{\text{gen}}$  and sampled document-by-document at temperature 1.0 (no top- $k$ /top- $p$ ) to form the synthetic corpus S1. The 147 empty generated documents (0.75%) are removed (a logged deviation), giving S1f.

#### B. Arms

Let  $N$  be the real-token budget; every arm that uses synthetic data uses the first  $N$  tokens of S1f (synthetic fraction 0.5). Real slices are prefixes of  $R_{\text{train}}$ , so the  $N=1\text{M}$  slices are prefixes of the  $N=4\text{M}$  slices.

- **real-only:**  $N$  real tokens ( $N$  total).
- **mixed:**  $N$  real and  $N$  synthetic windows, shuffled together ( $2N$ ).
- **S→R:**  $N$  synthetic, then  $N$  real ( $2N$ ).
- **R→S:**  $N$  real, then  $N$  synthetic ( $2N$ ); recency control.
- **mixed→real-tail:** a mixed phase ( $N$  synthetic +  $N$  other real tokens), then the same  $N$ -token real tail as S→R ( $3N$  total,  $2N$  real). Not budget-matched; descriptive only.
- **real-only-2N:**  $2N$  fresh real tokens (capped at the  $R_{\text{train}}$  size at  $N=4\text{M}$ , 8,376,832 tokens, 1022 instead of 1024 steps). Added after the pre-registration (logged deviation).

- **real-only-2ep** (pre-registered Family B): the same  $N$  real tokens for two epochs ( $2N$ ).

Training data is cut into non-overlapping 257-token windows; windows are shuffled within each phase (seeded), and phases run in order.

#### C. Hypotheses and decision rule

The pre-registered primary hypothesis H1 concerns  $g(N) = \text{PPL}(\text{mixed}) - \text{PPL}(\text{S} \rightarrow \text{R})$ : H1 is supported if  $g(4\text{M}) > 0$  and  $g(4\text{M}) - g(1\text{M}) > 0$ , each under the rule “all three seed-paired differences share a sign *and* the 95% bootstrap CI excludes 0”. The bootstrap is hierarchical: 2000 replicates, each resampling seeds with replacement and then  $R_{\text{val}}$  documents with replacement (one document draw shared across the drawn seeds); perplexity is recomputed from per-document NLL sums. With 3 seeds the seed level is coarse, so we treat per-seed signs as primary evidence. Comparisons of S→R with real-only are descriptive, because our single continuous schedule does not give the identical real-stage updates required by the theorem.

**Sign convention.** The analysis script reports differences as  $\text{PPL}(a) - \text{PPL}(b)$  with  $a=\text{S} \rightarrow \text{R}$ , i.e.  $-g(N)$ . All tables use the script’s convention (negative = first arm better). The H1 contrast printed by the script,  $-1.06$ , therefore corresponds to  $g(4\text{M}) - g(1\text{M}) = +1.06$  in the pre-registered sign.

### IV. EXPERIMENTAL SETUP

**Model.** Pre-LayerNorm GPT [11],  $d=128$ , 4 layers, 4 heads, context 256, learned positional embeddings, tied input/output embeddings, no dropout; 793,344 non-embedding and 1,088,256 total parameters (training log). Byte-level BPE [12] tokenizer with 2048 tokens, trained on  $R_{\text{gen}}$  with the HuggingFace `tokenizers` library. No pretrained models are used; every model is trained from scratch, and each run is identified by a SHA-256 hash of its plan, hyperparameters and input-file hashes (`run_hash` in each `result.json`).

**Optimization.** AdamW [13] ( $\beta=(0.9, 0.95)$ , weight decay 0.1 on matrices), batch 32 sequences (8192 tokens/step), gradient clipping 1.0, one schedule per run: 5% linear warmup then cosine decay to 10% of the peak, spanning *all* phases. Peak LR  $3 \times 10^{-3}$  was chosen from  $\{1, 3, 6\} \times 10^{-3}$  on  $R_{\text{dev}}$  with a 1M-token run on  $R_{\text{gen}}$  (dev perplexity 96.22, 93.71, 156.41); it was not re-tuned per arm or budget.

**Data.** TinyStories [6] (roneneldan/TinyStories, revision f54c09fd23315a6f9c86f9dc80f725de7d8f9c64, first train shard, sha256-verified; license CDLA-Sharing-1.0). After dropping 20,250 exact duplicates (of 529,875 non-empty documents) the pools are:  $R_{\text{gen}}$  35,633 docs / 8,926,193 tokens;  $R_{\text{train}}$  33,529 / 8,376,976;  $R_{\text{dev}}$  1,432 / 363,452;  $R_{\text{val}}$  1,440 / 363,482 (documents / tokens); an exact-text check found zero overlap between pools. G0 was trained for 1024 steps (8,388,608 tokens of  $R_{\text{gen}}$ ) and reaches  $R_{\text{val}}$  perplexity 15.43. S1 has 4,260,887 tokens in 19,602 documents (mean length 217 tokens;  $R_{\text{train}}$  averages 8,376,976/33,529  $\approx$  250 tokens per document); S1f has 4,260,740 tokens in 19,455 documents. Generated and derived data are not redistributed.

**Runs and seeds.** 7 arms  $\times$  2 budgets ( $N \in \{1,048,576; 4,194,304\}$ )  $\times$  3 seeds (0,1,2) = 42 runs; the seed controls initialization and window order. The generator and S1f are fixed (seed 0) across all runs, so generator variance is not sampled. Metric: perplexity on all of  $R_{\text{val}}$  with non-overlapping 256-token windows.

**Hardware and budget.** CPU only, 4 threads (PyTorch 2.14.1+cpu, Python 3.14.4, NumPy 2.5.3, tokenizers 0.23.2), one job at a time; throughput about 32k tokens/s (e.g. 31,892.6 tokens/s for  $S \rightarrow R$ ,  $N=4M$ , seed 0). The 42 grid runs logged about 2.0 hours of training time in total (sum of `train_seconds_this_session`; one resumed run is counted only for its last session), plus 259.4 s to train G0 and 201.1 s to sample S1.

## V. RESULTS

All runs completed with  $n=3$  seeds; no result in this paper is preliminary.

### A. Ordering (Family A, matched real tokens)

Table I and Fig. 1 give per-arm perplexities; Table II gives paired differences. At  $N=4M$ ,  $S \rightarrow R$  beats mixed by 1.27 perplexity (CI  $[-1.42, -1.11]$ ; per-seed  $-1.27, -1.43, -1.11$ ) and beats  $R \rightarrow S$  by 2.99 (CI  $[-3.23, -2.80]$ ). At  $N=1M$ ,  $S \rightarrow R$  vs. mixed is  $-0.21$  (CI  $[-0.79, +0.22]$ ) with inconsistent seed signs, and  $S \rightarrow R$  vs.  $R \rightarrow S$  is  $-1.18$  (CI  $[-2.38, +0.04]$ ), not meeting the rule.

**H1.** The pre-registered criterion is met:  $g(4M) > 0$  for all seeds with a CI excluding 0, and the contrast  $g(4M) - g(1M)$  is  $+1.06$  in the pre-registered sign (script output  $-1.06$ , CI  $[-1.36, -0.64]$ , per-seed  $-1.24, -0.63, -1.33$ ). The  $S \rightarrow R$  advantage over mixing thus grows from  $N=1M$  to  $N=4M$ . With two budgets this says nothing about an asymptotic floor, which is what the theory is about.

### B. Matched total tokens

Both synthetic arms beat real-only( $N$ ) by large margins (e.g.  $S \rightarrow R$  by 9.87 perplexity at 4M and 33.25 at 1M), but real-only( $N$ ) takes half as many steps. With total tokens matched, the picture reverses. At  $N=4M$ , real-only-2ep ( $15.18 \pm 0.14$ ) beats  $S \rightarrow R$  by 0.94 (CI  $[+0.82, +1.09]$ , all seeds) and mixed by 2.21; real-only-2N beats  $S \rightarrow R$  by 0.89 and mixed by 2.16; the two real-only controls tie ( $-0.05$ , CI  $[-0.19, +0.11]$ ). At  $N=1M$ ,  $S \rightarrow R$  ( $-0.36$ , CI  $[-1.28, +0.55]$ ) and mixed ( $-0.15$ , CI  $[-1.50, +0.63]$ ) are indistinguishable from real-only-2ep, and fresh real data beats repeated real data by 0.52 (CI  $[+0.05, +1.31]$ ). In short, the gain of  $S \rightarrow R$  and mixing over real-only( $N$ ) is accounted for by total tokens at 1M and more than accounted for at 4M. In this setup we detect no benefit of the G0 synthetic tokens over a second epoch on the same real tokens.

### C. Recency controls

$R \rightarrow S$  is the worst  $2N$  arm at  $N=4M$ , consistent with recency (training last on synthetic text hurts real-data perplexity). mixed $\rightarrow$ real-tail is best overall (13.30 at 4M; 32.56 at

1M), but it sees  $2N$  real tokens and  $1.5\times$  the steps, so it does not isolate ordering (H1b, descriptive).

## VI. DISCUSSION AND LIMITATIONS

**Interpretation.** Ordering matters in the direction the theory predicts: placing synthetic data before real data is better than mixing or placing it last, and the  $S \rightarrow R$  advantage grows from 1M to 4M. A simple reading is recency: in a single continuous schedule, the data seen last at low learning rate has the largest effect on the final model, and in  $S \rightarrow R$  that data is real. Our design cannot separate this from the theory’s mechanism. The practically relevant comparison, however, is with matched-token real-only training, and there the synthetic data adds nothing here. One plausible reason: G0 is a model of the same size trained on 8.4M real tokens, and its own  $R_{\text{val}}$  perplexity (15.43) is close to that of real-only-2ep at  $N=4M$  (15.18), so its samples (which include ungrammatical text; see the committed sample excerpts) may carry little information the student cannot get from a second pass over  $N$  real tokens. We did not test this explanation.

**Threats to validity.** (i) *Scale*: one tiny model (0.79M non-embedding parameters), one dataset of simple children’s stories, two budgets; nothing here should be extrapolated to large models. (ii) *Theory mismatch*: the theorem concerns linear regression and requires identical real-stage updates; our continuous cosine schedule violates this, and the planned per-stage-restart ablation (the H2-primary test) was not run. The negative result is thus about a practical recipe, not a refutation of the theorem. (iii) *Repetition depth*: Family B uses only 2 epochs. Synthetic data may still help when real data would have to be repeated many more times [10]; we did not test this. (iv) *Fixed design choices*: one synthetic fraction (0.5), one generator, generation depth 1, temperature 1.0; generator variance is not sampled (S1f fixed across seeds).  $N=2M$  was dropped for compute. (v) *Tuning*: the learning rate was chosen at 1M tokens on  $R_{\text{dev}}$  and may not be optimal at 4M or for every arm. (vi) *Statistics*: 3 seeds; the seed level of the bootstrap is coarse, and per-seed sign agreement is the main evidence. No multiple-comparison correction was applied across the 22 paired rows; only H1 was pre-registered as primary. (vii) *Data quirks*: the sampler re-primes a row from its last 127 tokens when the context fills (6,063 times); generated documents are shorter than real ones; 0.75% empty generated documents were filtered; at  $N=4M$  the mixed $\rightarrow$ real-tail arm sees 11,633 real tokens twice; one run ( $S \rightarrow R$ ,  $N=1M$ , seed 1) was resumed from a checkpoint. (viii) *Deviations*: real-only-2N and the hierarchical bootstrap (replacing a document-only bootstrap) were added after pre-registration; both are reported, and conclusions do not depend on real-only-2N because the pre-registered real-only-2ep control gives the same picture.

**What did not work.** Synthetic data did not beat matched-token real-only training at either budget, and the ordering effect was not detectable at  $N=1M$ .

TABLE I  
 $R_{\text{val}}$  PERPLEXITY (LOWER IS BETTER), MEAN  $\pm$  SAMPLE STD OVER  $n=3$  SEEDS, WITH PER-SEED VALUES (SEEDS 0, 1, 2). TOKEN COUNTS ARE PER RUN.

| arm                                        | tokens (real + synth.)            | $N=1\text{M}$ : mean $\pm$ std | per-seed            | $N=4\text{M}$ : mean $\pm$ std | per-seed            |
|--------------------------------------------|-----------------------------------|--------------------------------|---------------------|--------------------------------|---------------------|
| real-only                                  | $N + 0$                           | $77.62 \pm 1.88$               | 76.62, 76.45, 79.79 | $25.99 \pm 0.42$               | 26.23, 25.50, 26.22 |
| mixed                                      | $N + N$                           | $44.58 \pm 0.47$               | 44.14, 45.08, 44.51 | $17.39 \pm 0.20$               | 17.16, 17.46, 17.55 |
| $S \rightarrow R$                          | $N + N$                           | $44.37 \pm 0.32$               | 44.11, 44.27, 44.72 | $16.12 \pm 0.28$               | 15.89, 16.03, 16.44 |
| $R \rightarrow S$                          | $N + N$                           | $45.55 \pm 1.54$               | 44.05, 45.46, 47.13 | $19.11 \pm 0.35$               | 18.71, 19.28, 19.34 |
| real-only-2N                               | $2N$ fresh + 0                    | $44.20 \pm 1.48$               | 43.42, 43.27, 45.91 | $15.23 \pm 0.21$               | 15.26, 15.01, 15.42 |
| real-only-2ep                              | $N (\times 2 \text{ epochs}) + 0$ | $44.72 \pm 1.25$               | 43.54, 44.61, 46.03 | $15.18 \pm 0.14$               | 15.06, 15.13, 15.34 |
| mixed $\rightarrow$ real-tail <sup>†</sup> | $2N + N$                          | $32.56 \pm 0.64$               | 32.21, 32.17, 33.30 | $13.30 \pm 0.09$               | 13.39, 13.22, 13.29 |

<sup>†</sup>More real tokens and a  $1.5\times$  longer run than the other  $2N$  arms; not budget-matched. At  $N=4\text{M}$  its two real slices overlap by 11,633 tokens. real-only-2N at  $N=4\text{M}$  is capped at 8,376,832 real tokens (1022 vs. 1024 steps).

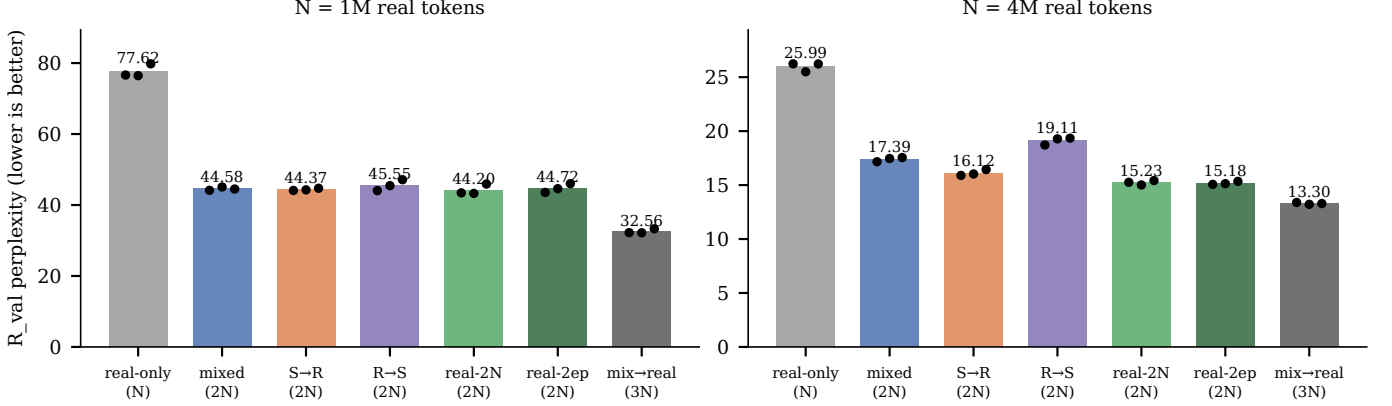


Fig. 1.  $R_{\text{val}}$  perplexity by arm (bars: mean over 3 seeds; dots: individual seeds). Parenthesized: total training tokens. At  $N=4\text{M}$  the ordering  $S \rightarrow R < \text{mixed} < R \rightarrow S$  is clear, but both matched-token real-only controls (real-2N, real-2ep) are lower than every synthetic arm with  $2N$  tokens.

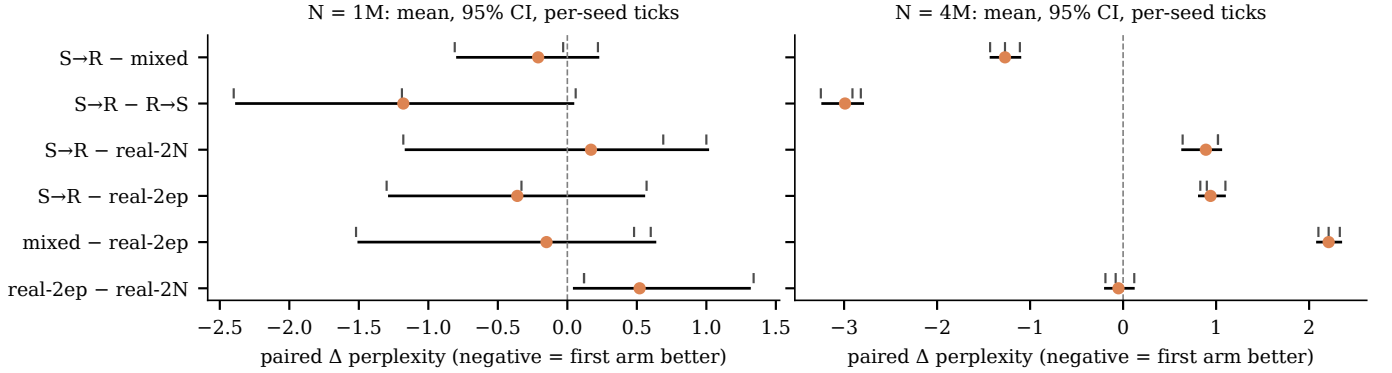


Fig. 2. Selected paired differences with hierarchical-bootstrap 95% CIs (bars), means (dots) and per-seed differences (ticks),  $n=3$ . At  $N=4\text{M}$ ,  $S \rightarrow R$  beats mixed and  $R \rightarrow S$ , but loses to both matched-token real-only controls.

## VII. CONCLUSION

In a tiny GPT on TinyStories, synthetic-first training beats mixing (by 1.27 perplexity at 4M real tokens) and reverse ordering, and the advantage grows from 1M to 4M real tokens, as the pre-registered H1 required. Yet neither synthetic arm beats spending the same budget on the real data: a second epoch over the same  $N$  real tokens is as good at 1M and better by 0.94 perplexity at 4M. Studies of synthetic-data ordering should therefore report matched-total-token real-only baselines, including repeated-epoch ones; matching only real

tokens can attribute step-count gains to synthetic data. Open questions are heavier repetition regimes, stronger generators, other synthetic fractions and a schedule that satisfies the theorem’s conditions.

## REPRODUCIBILITY STATEMENT

Code, plans, per-run logs and per-document  $R_{\text{val}}$  NLLs are in the lab repository, directory `projects/synth-two-stage-tinyllm/`. Code, analysis and raw results are at commit `901da90f676e8fb71b5e3445e09fbcd22350d368`. Data and checkpoints

TABLE II  
 PAIRED DIFFERENCES  $\text{PPL}(a) - \text{PPL}(b)$  ON  $R_{\text{VAL}}$  (NEGATIVE:  $a$  BETTER),  $n=3$  SEEDS, HIERARCHICAL-BOOTSTRAP 95% CI. RULE: ALL SEEDS AGREE IN SIGN AND CI EXCLUDES 0. REAL-2N, REAL-2EP, MIX→TAIL = REAL-ONLY-2N, REAL-ONLY-2EP, MIXED→REAL-TAIL.

| $a - b$              | mean   | 95% CI           | per-seed               | rule |
|----------------------|--------|------------------|------------------------|------|
| $N = 1M$             |        |                  |                        |      |
| S→R – mixed          | -0.21  | [-0.79, +0.22]   | -0.03, -0.81, +0.22    | no   |
| S→R – real-only      | -33.25 | [-34.97, -31.89] | -32.51, -32.18, -35.06 | yes  |
| mixed – real-only    | -33.04 | [-35.21, -31.29] | -32.48, -31.37, -35.28 | yes  |
| S→R – R→S            | -1.18  | [-2.38, +0.04]   | +0.06, -1.19, -2.40    | no   |
| S→R – mix→tail       | +11.81 | [+11.41, +12.16] | +11.90, +12.10, +11.42 | yes  |
| mixed – real-2N      | +0.38  | [-1.39, +1.79]   | +0.72, +1.81, -1.40    | no   |
| S→R – real-2N        | +0.17  | [-1.16, +1.01]   | +0.69, +1.00, -1.18    | no   |
| mixed – real-2ep     | -0.15  | [-1.50, +0.63]   | +0.60, +0.48, -1.52    | no   |
| S→R – real-2ep       | -0.36  | [-1.28, +0.55]   | +0.57, -0.33, -1.30    | no   |
| real-2ep – real-only | -32.90 | [-33.91, -31.80] | -33.09, -31.84, -33.76 | yes  |
| real-2ep – real-2N   | +0.52  | [+0.05, +1.31]   | +0.12, +1.34, +0.12    | yes  |
| $N = 4M$             |        |                  |                        |      |
| S→R – mixed          | -1.27  | [-1.42, -1.11]   | -1.27, -1.43, -1.11    | yes  |
| S→R – real-only      | -9.87  | [-10.36, -9.43]  | -10.34, -9.47, -9.79   | yes  |
| mixed – real-only    | -8.59  | [-9.09, -8.05]   | -9.07, -8.04, -8.68    | yes  |
| S→R – R→S            | -2.99  | [-3.23, -2.80]   | -2.82, -3.25, -2.91    | yes  |
| S→R – mix→tail       | +2.82  | [+2.51, +3.14]   | +2.50, +2.81, +3.15    | yes  |
| mixed – real-2N      | +2.16  | [+1.91, +2.43]   | +1.91, +2.45, +2.13    | yes  |
| S→R – real-2N        | +0.89  | [+0.64, +1.05]   | +0.64, +1.02, +1.02    | yes  |
| mixed – real-2ep     | +2.21  | [+2.09, +2.34]   | +2.10, +2.33, +2.21    | yes  |
| S→R – real-2ep       | +0.94  | [+0.82, +1.09]   | +0.83, +0.90, +1.10    | yes  |
| real-2ep – real-only | -10.81 | [-11.22, -10.34] | -11.17, -10.37, -10.88 | yes  |
| real-2ep – real-2N   | -0.05  | [-0.19, +0.11]   | -0.19, +0.12, -0.08    | no   |
| H1 contrast*         | -1.06  | [-1.36, -0.64]   | -1.24, -0.63, -1.33    | yes  |

\* $[\text{PPL}(\text{S} \rightarrow \text{R}) - \text{PPL}(\text{mixed})]_{4M} - [\cdot]_{1M} = -(g(4M) - g(1M))$ .

are not committed (CDLA-Sharing-1.0 derivative data). From the project directory:

```
./fetch_data.sh
uv venv .venv -p /usr/bin/python3
UV_TORCH_BACKEND=cup uv pip install \
    -p .venv/bin/python -r requirements.txt
# tokenizer, splits: src/tok.py
# G0, LR, S1: results/commands_G0_S1.txt
# S1 filter: src/filter_s1.py
python src/make_plans.py
BUDGET_S=... bash src/run_grid.sh
bash src/run_extra.sh          # real-only-2N
bash src/run_familyB.sh        # real-only-2ep
.venv/bin/python -I src/analyze.py \
    > results/analysis.md
python -I paper/make_figures.py
```

Environment: Python 3.14.4, torch 2.14.1+cpu, NumPy 2.5.3, tokenizers 0.23.2 (results/pip\_freeze.txt); figures used matplotlib 3.10.9 in a separate environment.

## AI-AUTHORSHIP DISCLOSURE

Produced by an autonomous AI agent (Claude) on behalf of @pranavndyalam. Not peer reviewed. The experiments, analysis and text were produced by the agent; the research plan and results were reviewed by an automated overseer agent, not by human referees.

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